

Automated IoT Smart Water Meter

Business Plan



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1 Problem Statement

The National Water Supply and Drainage Board (NWSDB) faces significant operational and financial challenges due to the continued reliance on conventional mechanical water metering systems. High Non-Revenue Water (NRW), inefficient manual processes, limited infrastructure visibility, and increasing maintenance costs reduce operational efficiency and negatively impact revenue generation. The key problems addressed by this project are discussed below.

1.1 High Non-Revenue Water and Revenue Loss

Non-Revenue Water (NRW) represents treated water that is produced but never billed due to physical leaks, unauthorized consumption, inaccurate metering, and administrative losses. According to the NWSDB, Sri Lanka records a national NRW rate between 24.8% and 27%, while Colombo historically experienced NRW levels approaching 48% before major rehabilitation initiatives.

The financial implications are substantial. NWSDB estimates that every 1% reduction in NRW generates approximately LKR 790 million in annual savings by reducing treatment, pumping, maintenance, and administrative costs. Consequently, reducing NRW remains one of the highest-impact opportunities for improving the financial sustainability of Sri Lanka's water sector.

1.2 Manual Meter Reading Inaccuracies

The existing billing process relies on manual meter reading performed by field personnel. This approach is time-consuming and susceptible to transcription errors, delayed billing cycles, estimated readings during inaccessible periods, and opportunities for meter tampering.

Furthermore, neither consumers nor utility operators have access to real-time water consumption data, limiting their ability to detect abnormal usage or identify leaks before significant water losses occur.

1.3 Lack of Remote Monitoring and Control

Current mechanical water meters do not provide remote communication or control capabilities. When pipe bursts, maintenance activities, or customer payment defaults occur, utility personnel must physically visit each property to inspect the meter or disconnect the water supply.

These manual interventions increase labour requirements, extend response times, and contribute to higher operational costs while delaying corrective actions.

1.4 High Maintenance Cost of Conventional Water Meters

Mechanical water meters gradually deteriorate through continuous operation, reducing measurement accuracy and increasing maintenance requirements. A JICA-supported pilot study conducted in the Harispattuwa Division in 2015 reported an 8.37% defective meter rate, resulting in approximately LKR 7.9 million in repair and maintenance costs within a single administrative division.

These findings demonstrate that conventional metering infrastructure imposes recurring maintenance expenses while limiting the ability of utilities to detect faults proactively.

1.5 Need for a Smart Water Metering Solution

The limitations of the existing water metering infrastructure highlight the need for a modern, intelligent metering system capable of providing accurate real-time consumption data, automated billing support, remote monitoring, leak detection, and improved operational efficiency. Such a solution has the potential to reduce Non-Revenue Water, lower maintenance costs, and enhance service delivery for both water utilities and consumers.

2 Stakeholder and Market Analysis

The success of the proposed Automated IoT Smart Water Meter depends on addressing the requirements of multiple stakeholders within Sri Lanka’s water supply ecosystem.

2.1 Stakeholders

The project serves three principal stakeholder groups, each with distinct operational requirements and expected benefits.

Table 1: Key Project Stakeholders and Their Needs

Stakeholder	Core Needs
NWSDB / Water Utilities	Real-time monitoring of NRW, automated leak detection, accurate meter readings, reliable billing data, reduced operation and maintenance costs, and improved network management.
Consumers / Households	Transparent water consumption information, accurate and dispute-free billing, mobile access to usage data, leak notifications, and alerts for remote disconnection or maintenance activities.
Government and Regulatory Authorities	Achievement of national NRW reduction targets, auditable metering data, improved resource utilization, reduced subsidy burden, and progress toward Sustainable Development Goal (SDG) 6: Clean Water and Sanitation.

2.2 Primary Customer

The primary customer for the proposed solution is the NWSDB, together with regional and provincial water boards responsible for public water distribution.

Unlike consumer-oriented products, the proposed business follows a Business-to-Government (B2G) and Business-to-Business (B2B) model, targeting a limited number of large institutional customers. This approach enables the company to focus its commercialization strategy on utility procurement processes, pilot deployments, regulatory approvals, and long-term service agreements rather than mass-market retail sales.

2.3 Market Validation

To validate market demand and refine product requirements, we conducted a stakeholder survey involving prospective users and utility representatives. The survey evaluated awareness of smart

water metering technologies, willingness to adopt the proposed solution, and key concerns related to implementation.

The findings indicated strong interest in smart metering, particularly for features such as real-time consumption monitoring, automated billing, leak detection, and mobile access to usage information. Respondents also identified installation cost, data privacy, and long-term reliability as important considerations for adoption.

3 Product Technology & Metrology

The Automated IoT Smart Water Meter developed combines low-power embedded hardware with wireless communication and cloud-based analytics to provide accurate, real-time water consumption monitoring. The system is designed for long-term deployment in residential water distribution networks while maintaining low manufacturing costs, minimal maintenance requirements, and extended battery life.

3.1 Hardware Stack Overview

The hardware architecture has been designed specifically for low-power, high-reliability operation in outdoor utility environments. Instead of employing costly ultrasonic flow sensors, the proposed design utilizes an efficient turbine-based flow measurement mechanism, significantly reducing manufacturing cost while maintaining the accuracy required for residential water metering.

The complete system consists of five major layers:

- **Flow Metering Unit:** A turbine rotor housed within a hydrodynamically optimized flow chamber.
- **Embedded Controller:** An STM32L0 ultra-low-power microcontroller responsible for pulse counting, flow computation, power management, and communication control.
- **Wireless Communication:** A SIM800L GSM/GPRS module that periodically transmits metering data to the cloud through existing cellular networks.
- **Cloud Backend:** A Supabase cloud database used for secure storage, synchronization, and analytics of meter data.
- **Software Platform:** WaterboardOS, a Software-as-a-Service (SaaS) dashboard that enables utilities to monitor consumption, detect anomalies, and manage customers remotely.

3.2 Flow Sensing and Computational Logic

Water entering the meter passes through a standardized cylindrical chamber containing an eight-blade turbine rotor. As the flow rate increases, the turbine rotates proportionally to the volume of water passing through the chamber.

Permanent magnets embedded within the turbine blades pass a waterproof Hall-effect sensor during rotation, generating a sequence of electrical pulses. The STM32L0 microcontroller counts these pulses and converts them into instantaneous flow rate and cumulative water consumption using calibration constants stored in memory.

The processed data are periodically transmitted via the GSM communication module to the cloud backend, where they are visualized through the WaterboardOS dashboard and the AquaSync consumer application. This architecture enables continuous remote monitoring, automated billing support, and early leak detection.

4 Business Model & Revenue Strategy

We adopt a Business-to-Government (B2G) and Business-to-Business (B2B) business model, targeting public water utilities as the primary customers. The business generates revenue through hardware sales, subscription-based software services, and future performance-based partnerships. This diversified revenue strategy ensures both immediate income from meter deployment and recurring revenue from cloud-based services.

4.1 Primary Business Model: B2G Hardware and Platform

The primary source of revenue is the sale of IoT smart water meters to the NWSDB and, in the future, regional and provincial water authorities. Each hardware deployment is bundled with access to the **WaterboardOS** cloud platform and the **AquaSync** consumer application.

A phased commercialization strategy is proposed:

- **Pilot Deployment**

A limited deployment of approximately 100–1,000 smart meters will be supplied to NWSDB at or near production cost. The objective of this phase is to validate field performance, collect operational data, and support product certification, regulatory approval, and long-term partnership agreements.

- **Large-Scale Deployment**

Following successful pilot validation, the system will be deployed on a larger scale. The expected selling price is approximately **LKR 18,500–25,000 per meter**, depending on production volume and hardware configuration. Future product versions may incorporate enhanced enclosure protection (IP67) and communication technologies such as LoRaWAN or NB-IoT for wider network coverage. Installation activities can be performed by NWSDB's existing field teams, minimizing additional operational costs.

- **Platform Subscription**

In addition to hardware sales, recurring revenue will be generated through subscription fees for the WaterboardOS platform. These subscriptions cover secure cloud hosting, data storage, software maintenance, analytics, technical support, and continuous platform updates. The software platform is designed to provide high availability with a target uptime of 99.9%, while ensuring secure access through JWT-based authentication and role-based data management.

4.2 Performance-Based Contracting

An alternative commercialization strategy involves performance-based contracting. Since NWSDB estimates that each one-percentage-point reduction in NRW saves approximately **LKR 790 million annually**, a gain-sharing model could be introduced in which our team receives a negotiated percentage of the verified financial savings generated through smart meter deployment.

This approach reduces the utility's upfront capital investment while aligning the incentives of both parties toward measurable reductions in water losses and improved operational efficiency.

4.3 Secondary Market Opportunities

Following successful validation with NWSDB, the proposed technology can be expanded into additional market segments.

Potential secondary customers include:

- Private condominium and apartment management companies requiring internal water sub-metering.
- Industrial and commercial facilities seeking automated water consumption monitoring.
- Hotels, hospitals, universities, and large institutional campuses.
- Regional water utilities in neighbouring countries experiencing similar NRW challenges.

The modular hardware architecture and cloud-based software platform enable the solution to be adapted for a wide range of utility-scale and commercial water management applications, supporting future regional expansion and long-term business growth.

5 Cost Estimation & Production Scaling

This section presents the estimated manufacturing cost of the prototype, the projected cost reductions achievable through large-scale production, and an illustrative business case demonstrating the economic viability of the proposed smart water metering solution.

5.1 Prototype Cost Estimation

The current prototype has been developed using commercially available development components suitable for testing and validation. As a result, the prototype cost is higher than the expected production cost.

Table 2: Estimated Prototype Cost (Single Unit)

Component	Cost (LKR)
STM32L0 microcontroller and supporting components	2,800
SIM800L GSM communication module	1,800
LiSoc12 Er34615	7800
GSM Load Switch AO3401A MOSFET	50
LDO Regulator XC6206	50
Boost Converter TPS61023DRLR	400
PCB fabrication, connectors, and miscellaneous components	5,000
enclosure	3,000
Total Prototype Cost	25,900

5.2 Production Scaling

Large-scale manufacturing significantly reduces the bill of materials (BOM) through bulk component procurement, optimized PCB fabrication, and streamlined assembly processes.

The projected production cost for a manufacturing volume of approximately 10,000 units is:

- Target manufacturing cost: **LKR 11,300 per unit**
- Bulk purchasing of electronic components to reduce overall BOM cost.
- Custom-designed PCB replacing development modules.
- Industrial-grade IP67 weatherproof enclosure for outdoor deployment.
- Migration from GSM-only communication to LoRaWAN or NB-IoT for improved scalability and lower communication costs.

- Production cloud infrastructure with 99.9% service availability, secure JWT-based authentication, and enterprise-grade data management.

The transition from prototype to commercial deployment will require collaboration with the NWSDB, including pilot installations, product validation, Memoranda of Understanding (MoUs), and formal type-approval certification.

5.3 Illustrative Business Case

The financial viability of the proposed solution is supported by the significant economic impact of reducing NRW.

Assuming a deployment of approximately 10,000 smart water meters, the estimated hardware investment would be:

$$10,000 \times \text{LKR } 11,300 = \text{LKR } 113,000,000$$

According to NWSDB estimates, every **1% reduction** in national Non-Revenue Water generates approximately **LKR 790 million** in annual savings through reduced treatment, pumping, maintenance, and administrative costs.

Therefore, even if a deployment of 10,000 smart meters contributes to only a fraction of a one-percentage-point reduction in NRW, the resulting financial savings have the potential to exceed the initial hardware investment within a relatively short period.

The selling price of the product was determined by considering the total production cost, desired profit margin, and market competitiveness. The proposed selling price is calculated using the following equation:

$$\text{Selling Price} = \text{Total Cost} + \text{Desired Profit}$$

The profit margin is calculated as:

$$\text{Profit Margin (\%)} = \frac{18,080 - 11,300}{18,080} \times 100$$

For the proposed product, the total manufacturing cost is estimated as Rs. 11,300, and the selected selling price is Rs. 25,000. Therefore, the profit margin is:

$$\text{Profit Margin (\%)} = \frac{25,000 - 11,300}{25,000} \times 100$$

$$= 54.8\%$$

Hence, the product can be sold at a price of Rs. 25,000, providing a profit margin of 54.8% while maintaining affordability and market competitiveness.

Although the exact return on investment will depend on deployment scale, installation costs, and network characteristics, this analysis demonstrates a strong economic justification for pilot deployment followed by large-scale implementation.

6 Conclusion

Our Automated IoT Smart Water Meter transforms a well-documented and financially significant national challenge, namely Non-Revenue Water (NRW) in Sri Lanka, into a practical engineering and commercial opportunity.

The proposed solution combines low-power hardware, reliable flow measurement, wireless communication, and cloud-based analytics to address the limitations of conventional mechanical water metering systems. With a functional prototype, a clearly defined bill of materials, and an established cost-reduction pathway for large-scale manufacturing, the project demonstrates strong potential for real-world deployment.

Furthermore, the development of the two-sided software ecosystem, consisting of **Water-boardOS** for utility management and **AquaSync** for consumer engagement, provides a complete smart water management platform rather than a standalone metering device. This integrated approach enables real-time monitoring, improved billing accuracy, leak detection, and enhanced operational decision-making.

The significant financial impact associated with reducing NRW strengthens the commercial feasibility of the proposed solution. NWSDB's own estimation of approximately **LKR 790 million in annual savings for every 1% reduction in NRW** provides a strong foundation for the business case and highlights the value of deploying intelligent metering infrastructure.

With further field validation, regulatory approval, and collaboration with NWSDB, our solution has the potential to progress from an academic prototype into a scalable utility technology capable of improving water management, reducing national water losses, and supporting sustainable infrastructure development in Sri Lanka.

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